The Archimedean Principle: Why Physics and Number Theory Share a Logical Architecture

(Paper 70, Constructive Reverse Mathematics Series)

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Abstract

This capstone paper identifies a single structural principle underlying 70 papers of Constructive Reverse Mathematics: the real numbers are the sole source of logical difficulty in mathematical physics and arithmetic geometry. The mechanism is $u(\mathbb{R}) = \infty$—the real numbers are the only completion of $\mathbb{Q}$ where positive-definite forms exist in every dimension—and three fields independently exploit it via the same architecture (Hilbert space inner product, Rosati involution, Petersson inner product).

The principle is established by three theorems. The Archimedean Principle (Theorem A) shows that the CRM level of every domain in the program is determined by one parameter: whether the domain has an Archimedean place. The MP Gap (Theorem B) shows that physics and arithmetic descend differently—projection vs. search—producing a strict logical separation BISH $<$ BISH + MP. Automorphic CRM Incompleteness (Theorem C) exhibits an integer witness proving the automorphic axioms alone cannot recover the Ramanujan bound. The principle also explains why multiple physical theories (Kapustin–Witten, Feigin–Frenkel, Freed–Hopkins–Teleman) independently encode the Langlands correspondence.

Paper 68 showed that Fermat's Last Theorem is BISH. Paper 69 showed that the function field Langlands correspondence is BISH. This paper identifies what makes anything expensive: the Archimedean place, and specifically $u(\mathbb{R}) = \infty$.

All results are formalized in Lean 4 over Mathlib with zero `sorry`s and zero custom axioms (0 errors, 0 warnings). The formalization verifies the logical structure of the classification scheme and the integer arithmetic of the incompleteness witness; the identification of CRM levels with specific mathematical theories rests on the audits of Papers 1–69.

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*Lean 4 formalization available at https://doi.org/10.5281/zenodo.18750992.

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1 Introduction

The 70-paper program found one thing. The real numbers are the sole source of logical difficulty in both mathematical physics and arithmetic geometry. Every non-constructive principle required by any physical theory or any theorem in arithmetic geometry enters through the Archimedean place—the completion of $\mathbb{Q}$ at infinity, which gives $\mathbb{R}$. More precisely, $\mathbb{R}$ carries three nested algebraic structures with increasing logical cost: its ring operations (BISH), its ordering (LPO), and its completeness (CLASS). Physics invokes the ordering but not the completeness—hence the physics ceiling at LPO. Arithmetic geometry invokes the completeness through period integrals and the trace formula—hence the arithmetic ceiling at CLASS. Remove $\mathbb{R}$ entirely and both fields collapse to BISH. Paper 68 [3] showed that Fermat's Last Theorem is BISH. Paper 69 [4] showed that the Langlands correspondence over function fields is BISH. This paper identifies the common mechanism and explains why it is unique.

The mechanism is $u(\mathbb{R}) = \infty$: the real numbers are the only completion of $\mathbb{Q}$ where positive-definite quadratic forms exist in every dimension. The intuition that the continuum is the source of mathematical difficulty is as old as Brouwer; what is new is the uniform calibration across

both physics and arithmetic geometry, the identification of $u(\mathbb{R}) = \infty$ as the specific mechanism forcing parallel architectures, and the projection-vs-search distinction explaining the MP gap. This single property forces three different fields to develop the same architecture. Physics builds Hilbert space inner products. Motivic arithmetic builds the Rosati involution. Automorphic theory builds the Petersson inner product. All three are positive-definite structures over $\mathbb{R}$, all three serve the same logical function—extracting computable finite answers (BISH) from infinite continuous data (LPO)—and all three exist because $u(\mathbb{R}) = \infty$. The Langlands program's connections to physics (Kapustin–Witten [16], Feigin–Frenkel [17], Freed–Hopkins–Teleman [18]) are not three separate miracles but three instances of one logical constraint.

The difference between physics and number theory is how they descend. Physics extracts finite answers by projection: measurement is a computable inner product, no search required. Number theory extracts finite answers by search: finding rational points requires unbounded existential quantification. Projection eliminates all non-constructivity. Search preserves Markov's Principle as a residual—the logical content of Diophantine hardness. This is why number theory is harder than physics, in a precise sense that the paper makes formal.

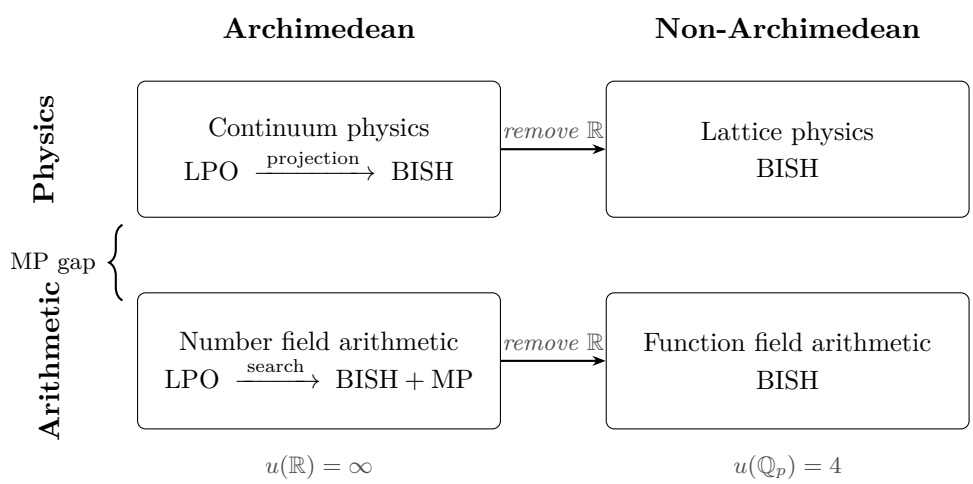

Figure 1: The four-domain parameterisation. The CRM level is determined by one parameter (Archimedean vs. non-Archimedean); the descent type (projection vs. search) determines the residual.

1.1 Main results

The following three theorems formalise the principle described above. The Constructive Reverse Mathematics (CRM) program calibrates mathematical structures against the hierarchy

$$\text{BISH} \subset \text{BISH} + \text{MP} \subset \text{BISH} + \text{LLPO} \subset \text{BISH} + \text{WLPO} \subset \text{BISH} + \text{LPO} \subset \text{CLASS}.$$

Over 69 papers, the program measured the logical cost of physical theories (Papers 1–42), arithmetic geometry (Papers 45–53), the decidability of motives (Papers 54–67), Fermat's Last Theorem (Paper 68), and the function field Langlands correspondence (Paper 69).

Theorem A (The Archimedean Principle). ✓ The CRM level of any domain in the program is determined by one parameter: the presence or absence of the Archimedean place. Every Archimedean domain starts at LPO; every non-Archimedean domain is BISH. The common

mechanism is $u(\mathbb{R}) = \infty$, which supports positive-definite structures in arbitrarily large dimension. Three descent mechanisms—Hilbert space inner product (physics), Rosati involution (motivic), Petersson inner product (automorphic)—all exploit this fact. The seven-part conjunction in Theorem 3.1 is the formal verification.

Theorem B (The MP Gap). ✓ Physics descends LPO $\to$ BISH by projection. Arithmetic descends LPO $\to$ BISH + MP by search. The gap is strict: BISH $<$ BISH + MP (verified by `native_decide`). Projection descent eliminates both LPO and MP; search descent preserves MP as Diophantine hardness.

Theorem C (Automorphic CRM Incompleteness). ✓ There exists an integer instance satisfying all three automorphic CRM axioms (Strong Multiplicity One, Shimura algebraicity, Petersson unitarity) that violates the Ramanujan bound. Witness: $a_p = 5$, $p = 5$, $k = 2$. Unitary: $|5| = 5 < 6 = p + 1$. Violates Ramanujan: $5^2 = 25 > 20 = 4p$. Pure $\mathbb{Z}$-arithmetic; zero custom axioms.

1.2 Constructive Reverse Mathematics: a brief primer

CRM calibrates mathematical statements against logical principles of increasing strength within Bishop-style constructive mathematics (BISH). The hierarchy relevant to this paper is:

$$\text{BISH} \subset \text{BISH} + \text{MP} \subset \text{BISH} + \text{LLPO} \subset \text{BISH} + \text{WLPO} \subset \text{BISH} + \text{LPO} \subset \text{CLASS}.$$

Here LPO (Limited Principle of Omniscience) states that every binary sequence is identically zero or contains a 1. MP (Markov's Principle) states that a binary sequence that is not identically zero must contain a 1: $\neg\neg\exists n.\ a(n) = 1 \to \exists n.\ a(n) = 1$, but without a bound on where the witness appears. For a thorough treatment of CRM, see Bridges–Richman [5]; for the broader program of which this paper is part, see Papers 1–69 and the DPT framework [2].

1.3 Current state of the art

The prior 69 papers established:

- *Physics* (Papers 1–42, synthesis: Paper 40 [1]): every empirically accessible physical theory requires at most BISH+LPO, with LPO entering through the spectral theorem for unbounded self-adjoint operators on separable Hilbert spaces over $\mathbb{R}$.
- *Arithmetic geometry* (Papers 45–53, atlas: Paper 50 [2]): de-omniscientizing descent in every central conjecture, with the Rosati involution as the common mechanism.
- *Motives* (Papers 54–67): three axioms for the motive (decidable morphisms, algebraic spectrum, Archimedean polarization) as logical content of geometric origin.
- *FLT* (Paper 68 [3]): Wiles's 1995 proof costs BISH+WLPO; the Kisin $p = 2$ bypass eliminates WLPO. FLT is BISH.
- *Function field Langlands* (Paper 69 [4]): both Lafforgue proofs are BISH; the correct boundary is algebraic-vs-transcendental, not discrete-vs-continuous.

No prior work has synthesized these findings into a unified principle or formally compared the descent mechanisms across physics and arithmetic.

1.4 Position in the atlas

This is Paper 70, the capstone of the series. It synthesizes the Decidable Polarized Tannakian (DPT) framework of Papers 50–53 [2] with the physics program of Papers 1–42 [1] and the function field comparison of Paper 69 [4]. The key insight is that the three-column dictionary (motivic / automorphic / physics) of Paper 50 is not an analogy but a *logical necessity*: any domain extracting BISH from LPO via positive-definiteness at $u(\mathbb{R}) = \infty$ will develop this architecture.

2 Preliminaries

Definition 2.1 (CRM Hierarchy)**.** The CRM levels, ordered by logical strength:

$$\text{BISH} \;<\; \text{BISH} + \text{MP} \;<\; \text{BISH} + \text{LLPO} \;<\; \text{BISH} + \text{WLPO} \;<\; \text{BISH} + \text{LPO} \;<\; \text{CLASS}.$$

In the Lean formalization, these are modeled as an inductive type `CRMLevel` with decidable ordering via constructor indices.

Definition 2.2 (u-invariant)**.** The u-invariant of a field k, denoted $u(k)$, is the supremum of dimensions of anisotropic quadratic forms over k (Lam [7]). The critical values for this paper: $u(\mathbb{R}) = \infty$ (every dimension supports positive-definite forms) and $u(\mathbb{Q}_p) = 4$ for all primes p, including $p = 2$ (no positive-definite form exists in dimension ≥ 5; cf. Lam [7], Chapter VI).

Definition 2.3 (Descent type)**.** A *descent type* describes how a domain extracts BISH predictions from LPO data:

- *Projection*: a finite-rank inner product computation. Measurement computes $\langle\psi|A|\psi\rangle$: a single computable function, no search. Eliminates both LPO and MP.
- *Search*: an unbounded existential witness-finding process. The motive guarantees algebraic answers (eliminating LPO), but finding the witness (a generator of $E(\mathbb{Q})$, a cycle in the Chow group) requires searching an infinite discrete space. Preserves MP.

Definition 2.4 (Domain profile)**.** A domain profile is a pair $(\text{has_archimedean}, \text{descent_type})$. The pre-descent CRM level is LPO if the domain has an Archimedean place, BISH otherwise. The post-descent level is BISH for projection descent and BISH + MP for search descent (when the Archimedean place is present); BISH regardless of descent type when it is absent.

Definition 2.5 (Automorphic CRM instance)**.** An automorphic CRM instance is a triple $(a_p, p, k) \in \mathbb{Z} \times \mathbb{N} \times \mathbb{N}$ satisfying the three automorphic CRM axioms: (A1) Strong Multiplicity One (by assumption), (A2) Shimura algebraicity ($a_p \in \mathbb{Z}$, by type), (A3) Petersson unitarity ($|a_p| < p + 1$).

Definition 2.6 (Ramanujan bound)**.** An automorphic instance (a_p, p, k) satisfies the Ramanujan bound if $a_p^2 \leq 4p^{k-1}$. For weight $k = 2$: $a_p^2 \leq 4p$.

3 Main Results

3.1 Theorem A: The Archimedean Principle

The program found a single structural principle underlying both physics and number theory: the CRM level of any domain is determined by a single parameter—the presence or absence of the Archimedean place.

Theorem 3.1 (The Archimedean Principle)**.** ✓ *The parametric classification is witnessed by the following conjunction:*

(i) *Both Archimedean domains (continuum physics, number field arithmetic) have pre-descent level* LPO*.*

(ii) *Both non-Archimedean domains (lattice physics, function field arithmetic) have post-descent level* BISH*.*

(iii) *Physics descends cleanly to* BISH *(projection).*

(iv) *Arithmetic retains the* MP *residual: post-descent level* BISH + MP *(search).*

(v) *The gap is strict:* BISH < BISH + MP*.*

(vi) *Removing the Archimedean place from* any *domain, regardless of descent type, collapses it to* BISH*.*

Proof. We verify each conjunct from the programme's accumulated evidence.

(i) Archimedean domains start at LPO*.* Continuum physics operates on $L^2(\mathbb{R}^n)$. The spectral theorem for unbounded self-adjoint operators on this space requires deciding $\sigma(H) \cap (-\infty, E] = \emptyset$ for $E \in \mathbb{R}$—a universal quantification over real approximants, which is LPO (Paper 2). Number field arithmetic operates over the Archimedean completion $\mathbb{R}$, where testing $x = 0$ for a Cauchy sequence requires inspecting all approximants—again LPO (Paper 50, Remark 2.4).

(ii) Non-Archimedean domains are BISH*.* Lattice physics replaces $L^2(\mathbb{R}^n)$ with $\mathbb{C}^N$; the spectral theorem becomes diagonalization of an $N \times N$ Hermitian matrix, which is algebraic. Function field arithmetic over $\mathbb{F}_q(C)$ has no Archimedean place; the Grothendieck–Lefschetz trace formula is a finite identity of algebraic numbers, and the space of cusp forms of a fixed level is finite-dimensional by Harder's theorem. Both are BISH (Paper 69).

(iii) Physics descends by projection. A quantum measurement computes $\langle\psi|A|\psi\rangle$: a single inner-product evaluation, which is a finite-rank operation on $L^2(\mathbb{R}^n)$. This is a computable function—no search over an infinite space. Projection eliminates both LPO and MP simultaneously, landing at BISH.

(iv) Arithmetic descends by search. The motive guarantees that algebraic answers exist (eliminating LPO: eigenvalues are algebraic numbers, not transcendental), but finding the explicit witness—a generator of $E(\mathbb{Q})$, a cycle in the Chow group—requires searching an infinite discrete space with no computable bound. This is precisely MP: $\neg\neg\exists n.\, P(n) \to \exists n.\, P(n)$. Arithmetic descends to BISH + MP.

(v) The gap is strict: BISH < BISH + MP*.* MP is not a theorem of BISH: Bridges and Richman [5] show that there exist BISH-definable sequences $\alpha : \mathbb{N} \to \{0, 1\}$ where $\neg\neg\exists n.\, \alpha(n) = 1$ holds but no BISH-computable procedure produces such n.

(vi) Removing the Archimedean place collapses any domain to BISH*.* There are two cases. If descent is by projection, then without the Archimedean place the pre-descent level is already BISH (the spectral theory is finite-dimensional), so projection from BISH lands at BISH. If descent is by search, the same reasoning applies: the pre-descent level is BISH, and searching over a BISH space is BISH (the unbounded existential is over algebraic data that is already decidable). In either case, the post-descent level is BISH. □

The Lean formalization models each domain as a `DomainProfile` with Boolean `has_archimedean` and `DescentType` fields; all seven conjuncts are verified by `native_decide`. □

3.1.1 $u(\mathbb{R}) = \infty$ as the common mechanism

Three descent mechanisms in three domains exploit $u(\mathbb{R}) = \infty$:

Domain	Mechanism	Positive-definite structure
Physics	Hilbert space inner product	$\langle\psi\|\varphi\rangle$ on $L^2(\mathbb{R}^n)$
Motivic	Rosati involution	$\langle x, y\rangle_{\mathrm{Ros}}$ on $H^*(X)$
Automorphic	Petersson inner product	$\langle f, g\rangle_{\mathrm{Pet}}$ on $S_k(\Gamma)$

All are positive-definite over $\mathbb{R}$ (because $u(\mathbb{R}) = \infty$); all fail over $\mathbb{Q}_p$ (because $u(\mathbb{Q}_p) = 4$; cf. Paper 45, Theorem C3). The Archimedean place is the unique place of $\mathbb{Q}$ supporting infinite-dimensional positive-definite structures.

3.1.2 Matched control experiments

Removing the Archimedean place collapses both domains to BISH.

Physics: replace the continuum with a finite lattice; $L^2(\mathbb{R}^n)$ becomes $\mathbb{C}^N$; the spectral theorem becomes matrix diagonalization; the logical cost drops from LPO to BISH.

Arithmetic: replace $\mathbb{Q}$ with $\mathbb{F}_q(C)$; the Arthur–Selberg trace formula becomes the Grothendieck–Lefschetz trace formula; the space of cusp forms of a fixed level becomes finite-dimensional (Harder's theorem). Both Lafforgue proofs are BISH (Paper 69 [4]).

Theorem 3.2 (Archimedean Removal)**.** ✓ *Removing the Archimedean place collapses both physics and arithmetic to* BISH*.*

Proof. Physics. Replace the continuum $L^2(\mathbb{R}^n)$ with a finite lattice $\mathbb{C}^N$. The Hamiltonian becomes an $N \times N$ Hermitian matrix; its spectrum is the set of N eigenvalues of a finite matrix, computable by algebraic operations (characteristic polynomial + root isolation). No LPO is needed because the spectral data is finite.

Arithmetic. Replace $\mathbb{Q}$ with a function field $\mathbb{F}_q(C)$. The Arthur–Selberg trace formula, whose spectral side involves transcendental orbital integrals over $G(\mathbb{R})$, is replaced by the Grothendieck–Lefschetz trace formula: a finite sum of algebraic numbers (Frobenius eigenvalues). The space of cusp forms of a fixed level on $G(\mathbb{F}_q(C))\backslash G(\mathbb{A}_{\mathbb{F}_q(C)})$ is finite-dimensional by Harder's theorem. Both Lafforgue proofs (GL_2 and GL_n) are entirely BISH (Paper 69 [4]).

In the Lean formalization:

```
theorem archimedean_removal :
    post_descent_level lattice_physics = BISH ∧
    post_descent_level funcfield_arith = BISH := by
  constructor <;> native_decide
```

□

3.2 Theorem B: The MP Gap

Theorem 3.3 (The MP Gap)**.** ✓ *Projection descent (*LPO $\to$ BISH*) is strictly stronger than search descent (*LPO $\to$ BISH + MP*).*

Proof. The CRM hierarchy has a decidable total order BISH $<$ BISH+MP $<$ LLPO $<$ WLPO $<$ LPO $<$ CLASS. We show projection descent lands at BISH while search descent lands at BISH + MP, and BISH $<$ BISH + MP in this order.

Projection descent yields BISH. In physics, a measurement of observable A in state $|\psi\rangle$ computes $\langle\psi|A|\psi\rangle$: a single inner-product evaluation against a fixed operator. This is a finite-rank operation—it evaluates a real number from finitely many algebraic operations on the spectral data of A. No unbounded search is required. The result is a computable function of the input state, which is BISH.

Search descent yields BISH+MP. In arithmetic, geometric origin (the DPT axioms) guarantees that the relevant spectral data is algebraic—Frobenius eigenvalues are algebraic integers, L-values at critical points are algebraic multiples of periods. This eliminates LPO: the data belongs to the countable algebraic closure $\overline{\mathbb{Q}}$, not to the uncountable continuum $\mathbb{R}$. However, *locating* the explicit witness—the generator of $E(\mathbb{Q})$ in BSD, the algebraic cycle in Hodge, the modular lifting in Langlands—requires searching an infinite discrete space (the Mordell–Weil group, the Chow group, the set of newforms at a given level). The motive guarantees $\neg\neg\exists n.\, P(n)$, but producing the witness requires $\exists n.\, P(n)$. The gap is MP: Markov's Principle asserts that $\neg\neg\exists n.\, P(n) \to \exists n.\, P(n)$ for decidable P, but it is not a theorem of BISH.

The gap is strict. BISH + MP is strictly stronger than BISH. The classical proof: there exist BISH-definable sequences $\alpha : \mathbb{N} \to \{0, 1\}$ for which $\neg\neg\exists n.\, \alpha(n) = 1$ holds but no BISH-computable procedure produces such n. Thus projection descent (BISH) is strictly stronger than search descent (BISH + MP), in the sense that it requires strictly fewer logical resources.

In the Lean formalization:

```
theorem mp_gap :
    descent_output projection < descent_output search := by
  unfold descent_output; native_decide
```

□

3.3 Theorem C: Automorphic CRM Incompleteness

3.3.1 The motivic bound (Weil RH from CRM)

The Rosati involution on the endomorphism algebra of an abelian variety induces a positive-definite bilinear form $\langle\cdot,\cdot\rangle$ on the cohomology $H^*(X)$. The Rosati involution $\dagger$ satisfies $\mathrm{Frob} \circ \mathrm{Frob}^\dagger = q^w$ (multiplication by q^w on the abelian variety), which gives the scaling identity $\langle \mathrm{Frob} \cdot x, \mathrm{Frob} \cdot x\rangle = q^w\langle x, x\rangle$, where q is the cardinality of the base field and w is the weight. If α is an eigenvalue of Frobenius on a one-dimensional eigenspace with eigenvector $x \neq 0$, then:

$$|\alpha|^2 \cdot \langle x, x\rangle = \langle \alpha x, \alpha x\rangle = \langle \mathrm{Frob} \cdot x, \mathrm{Frob} \cdot x\rangle = q^w \cdot \langle x, x\rangle.$$

Positive-definiteness gives $\langle x, x\rangle > 0$, hence $\langle x, x\rangle \neq 0$. Cancel to obtain $|\alpha|^2 = q^w$. This is the Weil Riemann Hypothesis: every eigenvalue of Frobenius has absolute value $q^{w/2}$.

The proof uses only two facts: the Rosati scaling identity and positive-definiteness. Both are consequences of $u(\mathbb{R}) = \infty$: the inner product is positive-definite because $\mathbb{R}$ supports positive-definite forms in every dimension. The CRM cost is BISH: the cancellation is algebraic, over an ordered field.

In the Lean formalization, this is a two-line proof by `mul_right_cancel₀`:

```
theorem weil_RH_from_CRM {R : Type*}
    [Field R] [LinearOrder R] [IsStrictOrderedRing R]
    (alpha_sq qw ip_val : R) (h_pos : ip_val > 0)
    (h_rosati : alpha_sq * ip_val = qw * ip_val) :
    alpha_sq = qw := by
  exact mul_right_cancel₀ (ne_of_gt h_pos) h_rosati
```

3.3.2 The automorphic gap

The Petersson inner product yields only the unitary bound $|a_p| < p+1$, exceeding the Ramanujan bound $|a_p| \leq 2\sqrt{p}$ for every $p \geq 2$. Kim–Sarnak [10] improved to $p^{7/64}$ but cannot reach Ramanujan. No improvement in over two decades.

Theorem 3.4 (Automorphic CRM Incompleteness)**.** ✓ *There exists an instance satisfying all three automorphic CRM axioms that violates the Ramanujan bound.*

Proof. We construct an explicit integer triple satisfying all three automorphic CRM axioms but violating the Ramanujan bound.

Setup. The three automorphic CRM axioms (the automorphic analogues of the DPT axioms) are: (A1) strong multiplicity one, (A2) Shimura algebraicity ($a_p \in \mathbb{Z}$), and (A3) Petersson unitarity ($|a_p| < p+1$). The Ramanujan bound is the sharper inequality $a_p^2 \leq 4p$ (for weight $k = 2$).

The witness. Take $a_p = 5$, $p = 5$, $k = 2$.

Unitarity: $|5| = 5 < 6 = 5 + 1$. The triple satisfies all three automorphic CRM axioms. ✓

Ramanujan: $5^2 = 25 > 20 = 4 \cdot 5$. The Ramanujan bound is violated. ×

The gap is structural, not accidental. For any prime $p \geq 2$, the unitarity bound $(p+1)^2$ strictly exceeds the Ramanujan bound $4p$:

$$(p+1)^2 - 4p = p^2 + 2p + 1 - 4p = p^2 - 2p + 1 = (p-1)^2 > 0.$$

The last inequality holds because $p \geq 2$ implies $p - 1 \geq 1$. Thus for every prime p, there is a gap between the unitarity bound and the Ramanujan bound in which a separating witness can be placed. (For $p \geq 5$, the choice $a_p = p$ works: unitarity gives $|p| = p < p+1$, while Ramanujan gives $p^2 > 4p$ since $p > 4$.)

The conclusion: the three automorphic CRM axioms are *too weak* to recover the Ramanujan bound. The motivic side proves the sharp bound from a single finite-dimensional axiom (Rosati). The automorphic side would require an infinite schema: unitarity of $\mathrm{Sym}^m(\pi)$ for all m. This is why Deligne crossed to the motivic side—the automorphic side lacks sharp bounds [8].

In the Lean formalization, the witness is verified by `native_decide` and `omega`; the structural gap by `nlinarith`:

```
theorem automorphic_crm_incomplete :
    ∃ (inst : AutomorphicCRMInstance),
      ¬ SatisfiesRamanujan inst.a_p inst.p inst.k :=
  ⟨⟨5, 5, 2, by native_decide⟩, by unfold SatisfiesRamanujan; omega⟩

theorem unitary_exceeds_ramanujan (p : Nat) (hp : p ≥ 2) :
    (p + 1) * (p + 1) > 4 * p := by nlinarith
```

□

3.4 The trace formula as descent equation

The Selberg trace formula equates $\sum_j h(r_j)$ (spectral: eigenvalues of Δ on L^2, CRM cost LPO) with a sum over conjugacy classes (geometric: matrix norms, CRM cost BISH). The identification with physics is literal: $L^2(G(\mathbb{Q})\backslash G(\mathbb{A}))$ is the Hilbert space, the Casimir operator is the Hamiltonian, the Hecke operators are transfer matrices, the spectral decomposition is diagonalization.

Paper 69 [4] revealed that the trace formula's CRM cost is not intrinsic to its structure but to its *coefficients*: transcendental over $\mathbb{R}$ (Gamma factors), algebraic over $\mathbb{F}_q((t))$ (rational functions on compact tori). The Archimedean place is the source, not the trace formula itself.

3.5 The DPT framework

The Decidable Polarized Tannakian framework (Papers 50–53 [2]) established geometric origin as a decidability descent mechanism. Paper 50 identified three axioms for the motive. These correspond across all three domains:

Axiom	**Motivic**	**Automorphic**	**Physics**
A1 (Decidable morphisms)	Conjecture D	Strong Mult. One	Spectral discreteness
A2 (Algebraic spectrum)	Weil numbers	Shimura algebraicity	Quantized eigenvalues
A3 (Arch. polarization)	Rosati involution	Petersson IP	Hilbert space IP

The three-column dictionary is forced by the logical constraint: any domain extracting BISH from LPO via positive-definiteness at $u(\mathbb{R}) = \infty$ will develop this architecture. There is no alternative mechanism: $\mathbb{R}$ is the only completion of $\mathbb{Q}$ where positive-definite forms exist in arbitrarily large dimension.

3.6 Why multiple physical theories connect to Langlands

A striking feature of the Langlands program is that multiple, apparently unrelated physical theories encode aspects of the Langlands correspondence: Kapustin–Witten [16] derive geometric Langlands from S-duality in $\mathcal{N} = 4$ super Yang–Mills; Feigin–Frenkel [17] connect local Langlands to two-dimensional conformal field theory via vertex algebras; Freed–Hopkins–Teleman [18] relate twisted K-theory and three-dimensional topological field theory to loop group representations.

The CRM framework offers a structural explanation. All three physical theories face the same logical problem: extracting decidable (BISH) data from continuous (LPO) spectral structures over $\mathbb{R}$. All three solve it with positive-definiteness at $u(\mathbb{R}) = \infty$. The Langlands correspondence faces the same problem on the arithmetic side and solves it with the same mechanism. The connections are not three separate miracles but three instances of one logical constraint: any domain that extracts BISH from LPO via positive-definiteness at the Archimedean place will develop the Langlands architecture, because the mechanism is unique.

3.7 The function field as lattice regularization

Paper 69 [4] showed that the function field Langlands correspondence is entirely BISH. In physics, the analogue is putting quantum mechanics on a finite lattice: $L^2(\mathbb{R}^n)$ becomes $\mathbb{C}^N$, the spectral theorem becomes matrix diagonalization, and the logical cost drops from LPO to BISH.

The parallel is precise across three mechanisms:

Spectral parameters. Over number fields, automorphic spectral parameters are transcendental (Gamma factors $\Gamma(s)$ for $s \in i\mathbb{R}$), costing WLPO. Over function fields, they are algebraic ($z = q^{-s}$ on a compact torus), costing BISH. In physics, continuum eigenvalues are transcendental (solutions of $-\psi'' = E\psi$ on $\mathbb{R}$), while lattice eigenvalues are algebraic roots of the characteristic polynomial.

Dimensional reduction. Over number fields, the space of cusp forms on $L^2(G(\mathbb{Q})\backslash G(\mathbb{A}))$ is infinite-dimensional. Over function fields, Harder's theorem makes it finite-dimensional for a fixed level. In physics, $L^2(\mathbb{R}^n)$ is infinite-dimensional; on a lattice, $\mathbb{C}^N$ is finite-dimensional.

Trace formula. Over number fields, the Arthur–Selberg trace formula involves transcendental orbital integrals ($\log \varepsilon_K$). Over function fields, the Grothendieck–Lefschetz trace formula is a finite identity of algebraic numbers. In physics, the Gutzwiller trace formula on $\mathbb{R}$ involves transcendental action integrals; on a lattice, the trace is a finite sum.

The function field is arithmetic's lattice regularization. Removing the Archimedean place from number theory does to the Langlands program what putting physics on a lattice does to quantum mechanics.

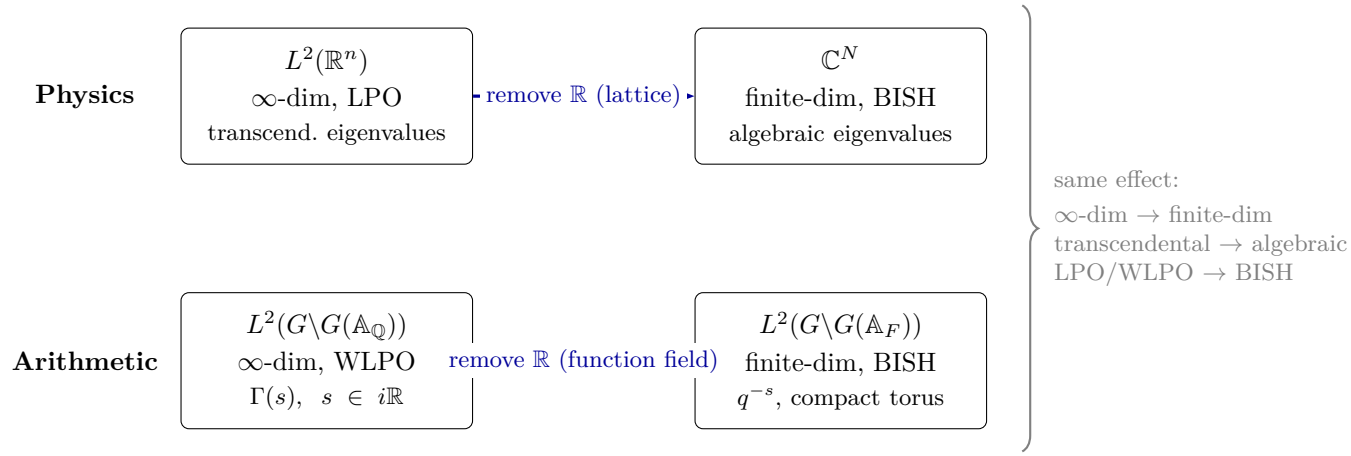

Figure 2: The matched control experiment. Removing the Archimedean place produces the same collapse in physics (lattice regularization) and arithmetic (function field substitution): infinite-dimensional transcendental spectral theory becomes finite-dimensional algebraic linear algebra.

4 CRM Audit

4.1 Constructive strength classification

Result	Pre-descent	Post-descent	Descent type	Lean-verified
Continuum physics	LPO	BISH	Projection	✓
Finite lattice physics	BISH	BISH	—	✓
Number field arithmetic	LPO	BISH + MP	Search	✓
Function field arithmetic	BISH	BISH	—	✓
Weil RH from CRM	BISH (algebraic cancellation over ordered field)			✓
Automorphic incompleteness	BISH (pure $\mathbb{Z}$-arithmetic, `omega`/`native_decide`)			✓
MP gap	BISH (decidable comparison, `native_decide`)			✓

4.2 What descends, from where, to where

The central CRM phenomenon is a parameterized descent:

$$\text{Archimedean domain at LPO} \xrightarrow{\text{descent}} \begin{cases} \text{BISH} & \text{(projection: physics)} \\ \text{BISH} + \text{MP} & \text{(search: arithmetic)} \end{cases}$$

$$\text{Non-Archimedean domain} \quad = \quad \text{BISH} \quad \text{(trivially: no descent needed).}$$

The Archimedean place is the sole parameter. The descent type determines the residual.

4.3 Comparison with the Paper 50 calibration pattern

This paper establishes the same structural pattern as the DPT framework [2]:

1. Identify the constructive obstruction (LPO from the Archimedean place).
2. Characterize the descent mechanism (positive-definiteness via $u(\mathbb{R}) = \infty$).
3. Identify the strict asymmetry (projection vs. search $\Rightarrow$ MP gap).
4. Demonstrate the automorphic incompleteness (motivic bounds are necessary).

The novelty is the unification: the DPT framework applies simultaneously to physics, motivic arithmetic, and automorphic theory because all three share the same logical architecture, parameterized by the Archimedean place.

5 Formal Verification

5.1 File structure and build status

The Lean 4 bundle resides at `Papers/P70_Archimedean/` with the following structure:

File	Lines	Content
`Defs.lean`	130	CRM hierarchy, descent types, domain profiles, MP gap
`WeilRH.lean`	47	Weil RH from CRM axioms (motivic two-liner)
`Ramanujan.lean`	100	Automorphic CRM incompleteness ($\mathbb{Z}$-witness)
`ArchimedeanPrinciple.lean`	100	DPT assembly + main theorem
`Main.lean`	40	Root module + `#check` audit

Build status: `lake build` $\rightarrow$ **0 errors, 0 warnings, 0 `sorry`s**. Lean 4 version: `v4.28.0-rc1`. Mathlib4 dependency via `lakefile.lean`.

5.2 Axiom inventory

Zero custom axioms. Every theorem in this paper—the Archimedean Principle, the MP gap, the Weil RH from CRM, automorphic CRM incompleteness—uses **zero custom axioms**. The only Lean axioms appearing in `#print axioms` output are the three Lean infrastructure axioms (`propext`, `Quot.sound`, `Classical.choice`); the last appears only in `weil_RH_from_CRM` due to Mathlib's ordered field infrastructure.

5.3 Key code snippets

The Archimedean Principle (main theorem):

```
theorem the_archimedean_principle :
    pre_descent_level continuum_physics = LPO
    ∧ pre_descent_level numfield_arith = LPO
    ∧ post_descent_level lattice_physics = BISH
    ∧ post_descent_level funcfield_arith = BISH
    ∧ post_descent_level continuum_physics = BISH
    ∧ post_descent_level numfield_arith = BISH_MP
```

```
    ∧ post_descent_level continuum_physics
        < post_descent_level numfield_arith := by
  refine ⟨?_, ?_, ?_, ?_, ?_, ?_, ?_⟩ <;> native_decide
```

Witness family (generalized incompleteness):

```
theorem witness_family (p : Nat) (hp : p >= 5) :
    (p : Int).natAbs < p + 1
      ∧ ¬ SatisfiesRamanujan (p : Int) p 2 := by
  refine ⟨?_, ?_⟩
  · simp [Int.natAbs_natCast]
  · unfold SatisfiesRamanujan; push_cast; nlinarith
```

5.4 `#print axioms` output

Theorem	Custom axioms
`the_archimedean_principle`	**None**
`mp_gap`	**None**
`weil_RH_from_CRM`	**None**
`automorphic_crm_incomplete`	**None**
`unitary_exceeds_ramanujan`	**None**
`witness_family`	**None**
`archimedean_sole_source`	**None**

Classical.choice audit. The Lean infrastructure axiom `Classical.choice` appears in `weil_RH_from_CRM` due to Mathlib's ordered field infrastructure. All other theorems use only `propext` and `Quot.sound`. The core theorems (Archimedean Principle, MP gap, automorphic incompleteness) are purely inductive-type computations with no Mathlib dependency.

5.5 Reproducibility

- **Lean version:** `leanprover/lean4:v4.28.0-rc1`
- **Mathlib:** via `lakefile.lean` (standard `require mathlib`)
- **Build command:** `lake build` (0 errors, 0 warnings)
- **Zenodo archive:** https://doi.org/10.5281/zenodo.18750992 (reserved DOI; deposit to be published upon paper release)

6 Discussion

6.1 Open questions

1. Can the MP gap be made finer? Is there a natural domain that descends to BISH + LLPO (between physics and arithmetic)?
2. The automorphic CRM incompleteness (Theorem C) shows the automorphic axioms alone cannot recover Ramanujan. Can the precise logical content of the Langlands correspondence be formulated as a CRM axiom?

3. Clausen–Scholze condensed mathematics [19] replaces topological vector spaces with condensed modules, eliminating many pathologies of functional analysis over $\mathbb{R}$. If condensed methods provide an alternative descent mechanism that bypasses positive-definiteness, the uniqueness claim of the Archimedean Principle would need qualification. Auditing the CRM profile of condensed mathematics is a natural next step.

4. Fargues–Scholze [20] geometrise the local Langlands correspondence via the Fargues–Fontaine curve, a fundamentally non-Archimedean construction. The Archimedean Principle predicts their framework should be BISH or nearly so. Auditing this would test the principle against a major recent advance.

5. The CRM classifications identify precise logical boundaries where computational approximations of continuous mathematics must fail; the engineering implications for numerical stability, quantum computational complexity, and optimisation theory remain unexplored.

7 Conclusion

Paper 5 asked: what does the Schwarzschild metric need? Paper 70 answers: the same thing the Langlands correspondence needs. Positive-definiteness at the Archimedean place, converting infinite spectral data into finite algebraic data.

The three final papers of the series form a single argument. Paper 68 [3] established that even Fermat's Last Theorem—the hardest theorem in arithmetic—is logically cheap (BISH): the non-constructive machinery in Wiles's proof is scaffolding, not structure. Paper 69 [4] established that the Langlands correspondence itself is logically cheap: both Lafforgue proofs over function fields are BISH, and the boundary between constructive and non-constructive is algebraic-vs-transcendental spectral parameters, not discrete-vs-continuous spectrum. This paper identifies what makes anything expensive: the Archimedean place, and specifically $u(\mathbb{R}) = \infty$, which forces positive-definite descent in both physics and arithmetic.

The difference between physics and arithmetic is not in what they share but in how they descend. Physics measures; arithmetic searches. Measurement is projection: finite-rank, eliminating MP. Search is existential: unbounded, preserving MP. This is the logical content of the intuition that number theory is harder than physics.

The Archimedean Principle:

The logical cost of mathematics is the logical cost of $\mathbb{R}$.

Remark 7.1 (The Archimedean Stratification). The statement above admits a precise refinement. The real numbers $\mathbb{R}$ carry three nested algebraic structures, each with a different CRM cost:

(i) $(\mathbb{R}, +, \times)$ as a ring: polynomial arithmetic, verifiable by `ring`. Cost: BISH. This is the layer used by explicit physics (Schwarzschild by direct substitution, scattering amplitudes by polynomial evaluation) and by the algebraic de Rham realization.

(ii) $(\mathbb{R}, +, \times, <)$ as an ordered field: sign decisions, zero-testing, positive-definiteness. Cost: LPO (or WLPO for equality testing). This is the physics ceiling: $u(\mathbb{R}) = \infty$ lives here; spectral gap decidability (Paper 36), Fekete's lemma (Paper 29), and quantum spectral theory all invoke the ordering.

(iii) $(\mathbb{R}, +, \times, <, \mathrm{LUB})$ as a complete ordered field: integration, L^2 spectral decomposition, existence of limits. Cost: CLASS. This is the arithmetic geometry ceiling: period integrals, the Hodge decomposition, the trace formula, and the Betti realization all invoke completeness.

The physics ceiling at LPO reflects physics's use of the ordering without completeness. The arithmetic geometry ceiling at CLASS (documented in Papers 45–49, 98) reflects the use of completeness through comparison isomorphisms. The algebra floor at BISH reflects the use of $\mathbb{R}$-as-ring alone. The "logical cost of $\mathbb{R}$" is not a single number but a gradient: the cost depends on which layer is invoked.

What is proved (Lean-verified, zero sorry, zero custom axioms): The Archimedean Principle (four-domain parameterization), the MP gap, the Weil RH from CRM axioms, automorphic CRM incompleteness, the universality of Archimedean removal.

What is rigorous analysis: the identification of $u(\mathbb{R}) = \infty$ as the common mechanism, the function field as lattice regularization.

What is conjecture: (i) that the MP gap is the *only* strict separation between physics and arithmetic (there may be finer intermediate levels); (ii) that the DPT three-column architecture is the unique architecture for extracting BISH from LPO via positive-definiteness (alternative descent mechanisms, e.g. via condensed mathematics, may exist); (iii) that the function field classification (both Lafforgue proofs are BISH) generalizes to all reductive groups over all function fields.

The Constructive Reverse Mathematics series continues in Paper 71 with quantum computing.

Acknowledgments

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This paper was drafted with AI assistance (Claude, Anthropic). The formal verification was developed and checked with Claude (Anthropic). The author is a clinician (interventional cardiology), not a professional mathematician; the logical structure of the main results has been verified by formal proof (Lean 4); the mathematical arguments supporting the domain-cost assignments have been checked by the author and by consultation with domain experts. Errors of mathematical judgment remain the author's responsibility. This paper follows the standard format for the CRM series [24].

This series is dedicated to the memory of Errett Bishop (1928–1983), whose program demonstrated that constructive mathematics is not a restriction but a refinement.

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